

Krushn Parmar

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EDUCATION

Vishwakarma Government Engineering College

Bachelor's in Engineering - Electronics & Communication

Ahmedabad, Gujrat

Aug. 2017 – May 2021

Alpha Vidhya Sankul

Higher Secondary Education

Junagadh, Gujarat

July 2015 – March 2017

EXPERIENCE

BLACKFIG Technologies

Software Development Engineer II

Bangalore, Karnataka

April 2024 - Present

- Familiar with assembly languages such as **PowerPC** and **x86_64**.
- Currently porting the GCC Compiler to VxWorks Workbench for the **AArch64 architecture**, enhancing toolchain support.

Software Development Engineer I

Jan. 2023 - March 2024

- Familiar with **Git** and **Jira** for efficient issue tracking and resolution.
- Worked extensively with **C++11**, **C++14**, and **C++17**, using core **OOP concepts** like **abstraction**, **inheritance**, and **polymorphism** to build practical applications.
- Familiar with **Linux concepts** like **multithreading** and **system calls**, with hands-on experience debugging and optimizing C/C++ code using **GDB** and **LLDB** and fixing bugs and in the **Dinkumware C++ library**.
- Experienced in using and contributing to the LLVM toolchain, including **Clang** and **LLVM**.
- Familiar with **LLVM tablegen DSL**.
- ICC-Integration:**
 - Ported the **ICC Compiler** to VxWorks Workbench enabling VxWorks Source Build and Image Project compilation with ICC for **X86_64 architecture**.
 - Performed comprehensive regression testing using DinkumWare Test Suite and benchmarked performance against Clang on Intel Super Micro Board.

Intern

Sep. 2022 - Dec. 2022

- Integrated the Eigen Library package into the VxWorks Workbench environment.
- Conducted testing of the Eigen Test Suite using vxTest Test Handler.
- Validated functionality on multiple architectures, including X86_64, PowerPC, RISC-V, and Aarch64, in both 32-bit and 64-bit modes.
- Performed tests on QEMU simulator and hardware platforms.

OPENSOURCE CONTRIBUTION

- PR77732** : [clang][PowerPC] Add flag to enable compatibility with GNU for complex arguments
- D152437** : Add phony registers for high halves of PowerPC/SPE SuperRegs
- D156351** : Fixed a regression in Clang (post-16.x) to accept -mabi=elfv1/elfv2 flags, resolving warnings and errors in Clang-trunk
- D156344** : Disable call to fma for soft-float
- D151711** : For PowerPC SPE add emergency stack spill slots

PROJECTS

Arsenic : | C++, CMake

<https://github.com/Long5hot/Arsenic>

- C++ implementation of the Lox interpreter.
- Implemented hand coded Lexer, Recursive decent parser and Tree-visitor, which supports control-flow, functions and basics of object-oriented language.

Mini libc++abi | C++

<https://github.com/Long5hot/libcplusplusabi>

- Mini version of libc++abi.
- Consists two-phase unwinding, search phase for Locate exception handlers by analyzing stack frames with LSDA and clean up phase to ensure resource cleanup through destructors during stack unwinding.

TECHNICAL SKILLS

Languages: C, C++ (C++11, C++14, C++17), Assembly (PowerPC, x86)

Developer Tools: Git, JIRA, CMake, Makefile, GDB, LLDB